

## LESSON 17

### PLANARITY AND EULER'S FORMULA

#### Overview:

This lesson will introduce basics of planar graphs and verifying them using Euler's formula.

#### Learning Outcomes:

At the end of Lesson 17, the students can:

1. determine whether a graph is planar or non-planar; and
2. verify a planar graph using Euler's formula.

#### Materials Needed:

- Modules (hard and soft copy)
- Devices (laptop and mobile phone)
- Social Media Platforms (Moodle, Google classroom, Facebook, YouTube and etc.).

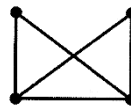
**Duration:** 1.5 hours

#### Learning Contents:

##### 17.1 Planar Graph

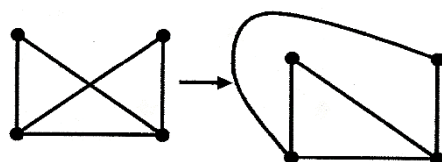
A **planar graph** is a graph that can be drawn so that no edges intersect each other (except at vertices). If the graph is drawn in such a way that no edges cross, we say that we have a **planar drawing** of the graph.

**EXAMPLE 17.1:** Is the planar drawing of a graph?  
Is the graph planar?

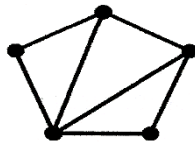


**Solution:**

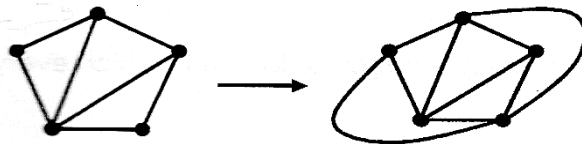
The drawing is not planar because two edges cross. However, the graph is planar because we can make an equivalent planar drawing of it as shown at the right.



**EXAMPLE 17.2:** Show that the graph on the right is planar.



Solution:

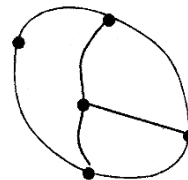


### 17.2 Euler’s Formula

In a connected planar graph drawn with no interesting edges, let  $v$  be the number of vertices,  $e$  the number of edges, and  $f$  the number of faces. Then  $v + f = e + 2$ .

Note that in a planar drawing of graph, the edges divide the graph into different regions called **faces**. The region surrounding the graph, or the exterior, is also considered as a face, **infinite face**.

**EXAMPLE 17.3:** Count the number of edges, vertices and faces in the planar below, and then verify Euler’s formula.

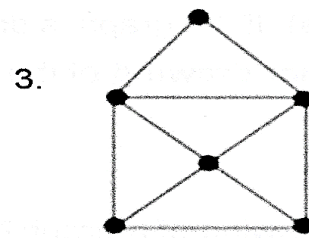
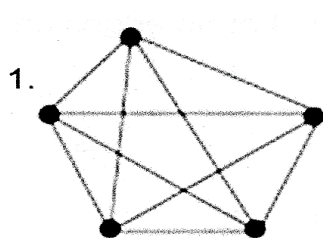


Solution:

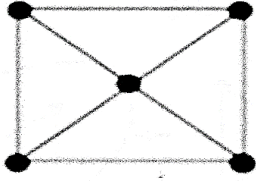
$$\begin{aligned} \text{Given: } v &= 5, e = 7, \text{ and } f = 4 \\ v + f &= e + 2 \\ 5 + 4 &= 7 + 2 \\ 9 &= 9 \end{aligned}$$

### Learning Activity:

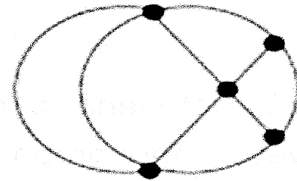
- 1) Show that the graph is a planar by finding a planar drawing and verify Euler’s formula in the found planar graph.



2.



4.



### Learning Evaluation:

- 1) Sketch a planar graph (without multiple edges or loops) in which every vertex has 3 degree.
- 2) Sketch a planar graph (without multiple edges or loops) in which every vertex has 4 degree.

### References:

- Aufman, R. et. Al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.
- Commission on Higher Education. (2016). *General Mathematics [Teaching Guide]*. Diliman, Quezon City. Retrieved November 15, 2021 from <https://www.academia.edu/29721020/>